



DO-IT-YOURSELF: PROJECT

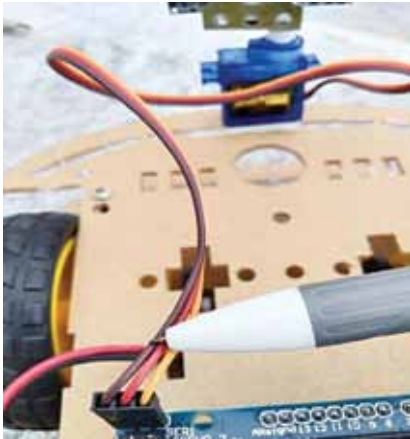


Fig. 10: Servo setup

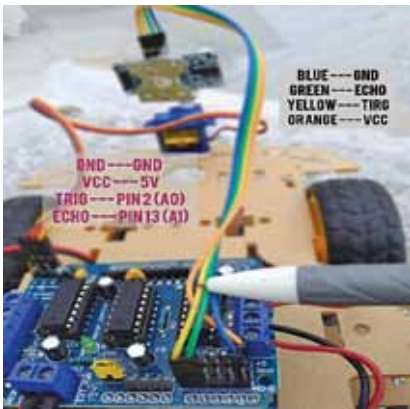


Fig. 11: Sensor connections

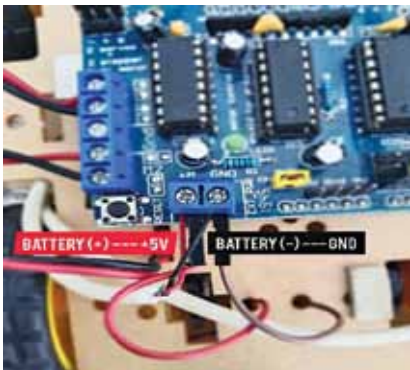


Fig. 12: Power supply connections

the bottom of the chassis

- Attach the wheels to the DC motors

Step 2. *Mounting the components:*

- Screw the Arduino onto top chassis
- Then plug the motor driver shield into Arduino
- Connect the motors to correct pair of terminals—the front motors to front terminals of motor shield and the rear motors to rear terminals of motor shield

Step 3. *Setting up the servo:*

- Attach the servo on top of the chassis
- Plug the servo into Servo 1 (SER1) header pins on the shield

Step 4. *Preparing the ultrasonic sensor: (Fig. 11: Sensor connections)*

Place the ultrasonic sensor on the servo using the sensor stand. Then plug four wires into the sensor using female jumpers. Next, connect sensor to Arduino Uno as described below:

GND→GND
Vcc→5V
TRIG→pin 2 (AO)
ECHO→pin 13 (A1)

Step 5. *Powering the robot:*

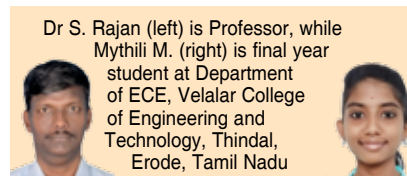
Connect the Lipo battery to the L293D motor driver as follows:

Lipo battery (+)→+12V
Lipo battery (-)→GND

The obstacle-avoiding robot is now ready! It can be used for such applications as:

- Mobile navigation systems, military and law enforcement, search and rescue, and forest conservation.
- For household work like automatic vacuum cleaning.
- With proper programming, for weightlifting and auto parking assistance. **EFY**

EFY Note
The source code of this project is available for free download at www.electronicsforu.com



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